

**SECOND SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

MEC 2C 02—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 15

Maximum : 15 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 2C 02—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SIDE Candidates)

1. A system with m equations and n variables has at most _____ basic solutions.
(A) $n - m$. (B) $m - n$.
(C) Cm . (D) None of these.
2. A basic feasible solution is a basic solution whose variables are _____.
(A) Feasible. (B) Negative.
(C) Non-negative. (D) None.
3. The line joining (0, 0) to (100,100) in a Lorenz Curve is called :
(A) Line of equal distribution. (B) Sloping line.
(C) Line of perfect inequality. (D) None of these.
4. The divergence between Lorenz curve and line of perfect equality can be measured by :
(A) Gini co-efficient. (B) Co-efficient of variation.
(C) Both. (D) None.
5. In LPP, simplex method was developed by :
(A) Koopman. (B) G.B. Dantzig.
(C) Leontief. (D) None of these.
6. Any non negative value of (x_1, x_2) is a constraints :
(A) Critical region. (B) Feasible region.
(C) Optimal solution. (D) None.
7. The optimal solution any LPP corresponds to one of the _____ of the feasible region.
(A) Turning points. (B) Corner points.
(C) Maximum point. (D) None.

8. In LPP we deal with _____ objectives.
- (A) Many. (B) Two.
(C) Three. (D) Four.
9. The effect of changes in the co-efficient in the optimum value of the objective function can be studied through a technique called :
- (A) Simplex method. (B) Sentitivity analysis.
(C) Assignment problem. (D) None of these.
10. Dual of the dual is called :
- (A) Simplex. (B) Primal.
(C) Both. (D) None.
11. When both players use their optimal minimax strategies the resulting expected pay off is called :
- (A) Game. (B) Strategy.
(C) Both. (D) None.
12. In a game, minimax value is _____ maximin value.
- (A) Greater than. (B) Geater than or equal to.
(C) Less than. (D) None.
13. Saddle point is also called :
- (A) Critical point. (B) Equilitarian point.
(C) Both. (D) None.
14. The input-output analysis technique was propounded by :
- (A) Van Neumann. (B) W.W. Leontief.
(C) C.F. Christ. (D) None.
15. An input-output model in which some of the production is consumed by external bodies is called :
- (A) Open. (B) Closed.
(C) Both (A) and (B). (D) None.

**SECOND SEMESTER (CBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

MEC 2C 02—MATHEMATICAL ECONOMICS

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. Define a Lorenz curve.
2. What is meant by Line of best fit ?
3. Explain Closed input output model.
4. What is an Endogenous variable ?
5. State the Chain rule.
6. What is an Objective function ?
7. State the Law of diminishing marginal productivity.
8. Define Jacobian derivative.
9. What is meant by Unconstrained optimization ?
10. What is meant by a Saddle point ?
11. What is meant by Technological coefficient matrix ?
12. Explain the term Nondegenerate constraint qualification.

(8 × 3 = 24 marks)

Turn over

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Explain the measurement of income inequality using Lorenz curve.
14. Explain the method of least squares.
15. Examine whether the input-output system with the following co-efficient matrix is feasible :

$$\begin{bmatrix} 0.5 & 0.6 \\ 0.3 & 0.7 \end{bmatrix}.$$

16. Derive the profit maximizing conditions for a discriminating monopolist selling in two markets.
17. Explain the Hessian Matrix.
18. Explain the limitations of input-output analysis
19. Describe the Kuhn-Tucker formulation for a constrained minimization problem.

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. Explain the conditions for profit maximization of a firm. If the cost function of a firm is given by $C = 25x + 10000$ and the demand function is given as $p = -1/5x + 500$, find the profit maximizing level of output.
21. Explain the basic assumptions of input-output analysis. The final demand of two sectors is given as $X_2a = \text{Rs. 20 crores}$ and $X_2b = \text{Rs. 15 crores}$. The co-efficient matrix is given as

$$C = \begin{bmatrix} 0.4 & 0.1 \\ 0.2 & 0.3 \end{bmatrix}. \text{ Find the levels of output } X_1 \text{ and } X_2.$$

(1 × 11 = 11 marks)